

Module 3



25-200-0106A ENVIRONMENTAL AND LIFE SCIENCES

Dr. JOB THOMAS
Dean, Faculty of
Environmental Science,
Cochin University of
Science and Technology

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COURSE OUTCOMES

1. Identify the global environmental issues
2. Examine the types of pollution in society along with their sources
3. Elucidate the basic biological concepts via relevant industrial applications and case studies.
4. Evaluate the principles of design and development, for exploring novel bioengineering projects



MODULE 3

- Biomolecules and their Applications (Qualitative): Carbohydrates (cellulose-based water filters, PHA and PLA as bioplastics), Nucleic acids (DNA Vaccine for Rabies and RNA vaccines for Covid19, Forensics – DNA fingerprinting), Proteins (Proteins as food – whey protein and meat analogs, Plant based proteins), Lipids (biodiesel, cleaning agents/detergents), Enzymes (glucose-oxidase in biosensors, lignolytic enzyme in bio-bleaching).

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MODULE 3 (COND....)

- Nature-Bioinspired Materials and Mechanisms (Qualitative): Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Shark skin (Friction reducing swim suits), Kingfisher beak (Bullet train). Human Blood substitutes - hemoglobin-based oxygen carriers (HBOCs) and perflourocarbons (PFCs).

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BIOMOLECULES

- Biomolecules, or biological molecules, are organic compounds produced by living organisms that are essential for life's processes, providing structural support and carrying out key functions
- Examples:
 - carbohydrates
 - lipids
 - proteins
 - nucleic acids
 - enzymes

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CARBOHYDRATES

- A carbohydrate is an organic biomolecule made of carbon, hydrogen, and oxygen, often with a 2:1 hydrogen-to-oxygen ratio, and serves as the primary energy source and a structural component in living organisms
- Applications:
 - Cellulose-based water filters
 - PHA and PLA as bioplastics for packaging

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CELLULOSE-BASED WATER FILTERS

- Filters made from cellulose, the abundant carbohydrate found in plant cell walls
- The material traps particles, pathogens, and other contaminants as water passes through it.
- Benefits:
 - Environmentally friendly and biodegradable.
 - Made from renewable resources.
 - High porosity, large surface area, and good mechanical strength.
 - Affordable and versatile for household, industrial, and agricultural use.

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PHA AND PLA AS BIOPLASTICS

- Produced from corn, sugarcane, or cassava and agricultural waste.
- PHA (Polyhydroxyalkanoates): Packaging, disposable items, and even medical devices.
- PLA (Polylactic Acid): Widely used in the packaging industry for food and beverage containers and also in medical applications.

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BIOPLASTIC

- Benefits:
 - Biodegradable alternatives to conventional plastics.
 - Reduced reliance on fossil fuels.
 - Good processability and mechanical properties comparable to conventional plastics

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NUCLEIC ACIDS

- Nucleic acids are vital biomolecules found in all living cells and viruses, serving to store, transmit, and express genetic information.
- The two primary types are
 - Deoxyribonucleic Acid (DNA)
 - Ribonucleic Acid (RNA)

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DNA VACCINE FOR RABIES

- DNA vaccines for rabies are a promising alternative to traditional vaccines, particularly for animal vaccination campaigns in developing countries.
- Though they have shown great potential in animal models, their application for human use is more complex and faces hurdles.

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RNA VACCINES FOR COVID19

- COVID-19 RNA vaccines, also known as messenger RNA (mRNA) vaccines, were groundbreaking in the fight against the pandemic, delivering genetic instructions to your cells to create a key viral protein and train your immune system.

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FORENSIC DNA FINGERPRINTING

- Forensic DNA fingerprinting, or DNA profiling, is a technique for identifying individuals by analyzing the unique patterns of their DNA, primarily using Variable Number of Tandem Repeats (VNTRs).
- It involves DNA extraction, restriction enzyme digestion, and gel electrophoresis to separate DNA fragments, with the resulting pattern used for comparisons in criminal investigations, paternity testing, and human remains identification

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PROTEINS

- Proteins are essential biomolecules composed of chains of amino acids, performing vital functions in all living organisms, from building and repairing cells and tissues to catalyzing reactions as enzymes, transporting molecules, and regulating bodily processes

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PROTEINS AS FOOD

- Protein comes from both animal sources, such as meat, poultry, fish, eggs, and dairy, and plant-based sources, including beans, lentils, nuts, seeds, and soy products.
- These foods are crucial for growth, repair, and overall health, providing essential nutrients like vitamins, minerals, and fiber.

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ANIMAL-BASED PROTEIN SOURCES

- Meat and Poultry: Choose lean cuts of beef, pork, and chicken or turkey.
- Fish and Seafood: Opt for fish rich in omega-3 fatty acids, such as salmon, trout, and anchovies.
- Eggs: A versatile and easily available source of protein.
- Dairy Products: Milk, yogurt (especially Greek yogurt), and cheese (like cottage cheese) are good options.

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PLANT-BASED PROTEIN SOURCES

- Legumes: Beans, peas, and lentils (such as chickpeas, kidney beans, and split peas) are low in fat and high in fiber and protein.
- Nuts and Seeds: Almonds, walnuts, chia seeds, and pumpkin seeds provide protein and healthy fats.
- Soy Products: Edamame, tofu, and tempeh are excellent plant-based protein sources.
- Whole Grains: Some grains, including quinoa, oats, and buckwheat, also contribute to protein intake.

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WHEY PROTEIN AND MEAT ANALOGUE

- To replicate the appearance, texture, and flavor of meat products using alternative protein sources
- Whey protein is a milk-derived protein known for its gelling and binding properties, which are crucial for building the three-dimensional structure of a meat analogue.
- While many meat analogues rely on plant proteins, hybrid products can be developed using a combination of plant proteins and animal-derived proteins like whey to achieve desired sensory and nutritional characteristics.

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LIPIDS

- Lipids are a diverse group of water-insoluble organic compounds, including fats, oils, waxes, sterols, and fat-soluble vitamins.

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BIODIESEL FROM LIPIDS

- Lipids serve as an excellent renewable and sustainable source for producing biodiesel through a chemical process called transesterification
- fats and oils react with an alcohol (like methanol) to create biodiesel and glycerol.
- lipid sources are vegetable oils, animal fats, used cooking oil, and lipids from oleaginous microorganisms (like yeasts and microalgae) or even municipal sludge
- Limitation is economic viability

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CLEANING AGENTS/DETERGENTS FROM LIPIDS

- Cleaning agents derived from lipids are primarily soaps and detergents, created through the saponification process, where fats and oils (lipids) are reacted with an alkali.
- Saponification is the chemical process that transforms a fat or oil (a lipid) into soap and glycerol (an alcohol) by reacting it with a strong alkali, such as sodium hydroxide or potassium hydroxide

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ENZYMES

- Enzymes are biological catalysts, most often proteins, that accelerate chemical reactions in living organisms by lowering the activation energy required for the reaction to occur.
- They are highly specific, binding to molecules called substrates at their active sites and converting them into products without being consumed by the reaction.

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GLUCOSE-OXIDASE IN BIOSENSORS

- A glucose oxidase biosensor is an electrochemical device that uses the enzyme glucose oxidase (GOx) to measure glucose levels in biological samples.
- GOx converts glucose into gluconic acid and hydrogen peroxide, with the hydrogen peroxide then detected through electrochemical means (like oxidation to oxygen), which allows for the quantification of glucose.
- These biosensors are used for diabetes management

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LIGNINOLYTIC ENZYMES IN BIOLEACHING

- Bioleaching is a process in mining and extracting valuable metals from a low-grade ore with the help of microorganisms such as bacteria.
- Ligninolytic enzymes produced by bacteria are used in bioleaching to break down recalcitrant lignin in lignocellulosic materials, making metal ions or other compounds more accessible.
- examples for Ligninolytic enzymes are laccases, lignin peroxidases (LiP), and manganese peroxidases (MnP),

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NATURE-BIOINSPIRED MATERIALS AND MECHANISMS

- Echolocation
- Photosynthesis
- Bird flying
- Lotus leaf effect
- Plant burrs
- Shark skin
- Kingfisher beak

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ECHOLOCATION

- Locating an objects by reflected sound, that used by animals such as dolphins and bats
- Examples
 - Ultrasonography
 - sonars

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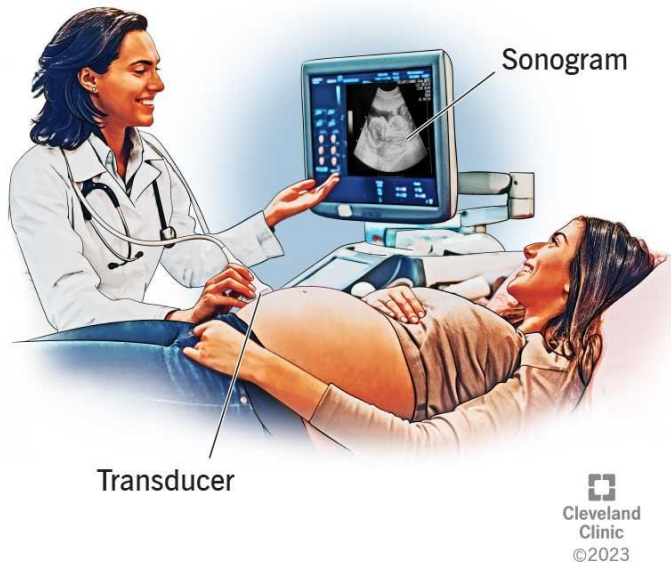
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Abdominal ultrasound



ULTRASONOGRAPHY



- Ultrasonography, or ultrasound, is a medical imaging technique that uses high-frequency sound waves to create real-time images of organs, tissues, and other structures inside the body, allowing doctors to diagnose various conditions without surgery.

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ULTRASONOGRAPHY

- A handheld device called a transducer emits sound waves that reflect off internal structures, creating echoes that a computer converts into images called sonograms displayed on a screen.
- Common applications include monitoring fetal development during pregnancy, detecting tumors, and examining organs like the liver, heart, and kidneys

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SONAR

- Sonar, an acronym for SOund NAvigation and Rangin
- It is a technology that uses sound waves to detect, navigate, and communicate underwater by sending out sound waves and listening for echoes that bounce off objects, or by simply listening for sounds made by objects.

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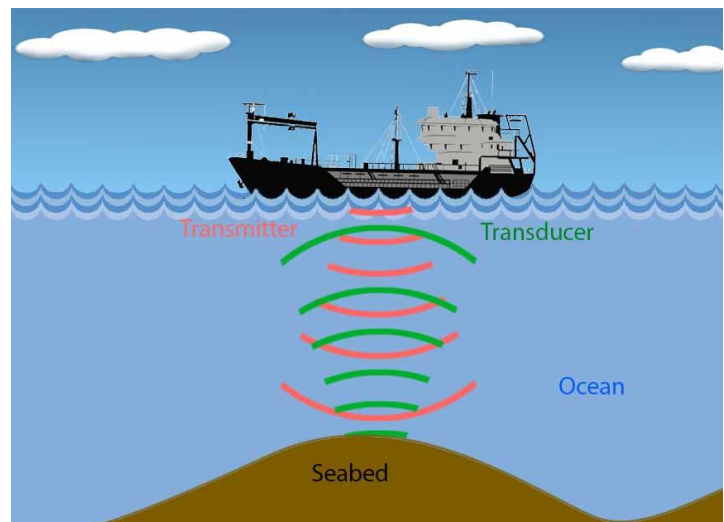
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SONAR

- Sound waves travel farther in water than light or radar waves, making sonar ideal for exploring the ocean and mapping the seafloor.



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PHOTOSYNTHESIS

- Photosynthesis is a fundamental biological process where green plants, algae, and some bacteria use sunlight, water, and carbon dioxide to create their own food (sugars) and release oxygen as a byproduct

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PHOTOVOLTAIC CELLS

- Photovoltaic (PV) cells, also known as solar cells, convert sunlight directly into electricity, a process similar to photosynthesis's conversion of light into chemical energy in plants.
- While photosynthesis uses chlorophyll and chloroplasts to capture light and create biofuels and biomass, PV cells use semiconductor materials to absorb photons and dislodge electrons, generating an electrical current

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BIONIC LEAF

- A "bionic leaf" mimics natural photosynthesis by using a combination of artificial components and living organisms to convert sunlight, water, and carbon dioxide into fuels or other chemicals
- Inspired by a natural leaf, it uses solar energy to split water into hydrogen and oxygen, which is then consumed by genetically engineered bacteria to produce liquid fuels like isopropanol, or other products

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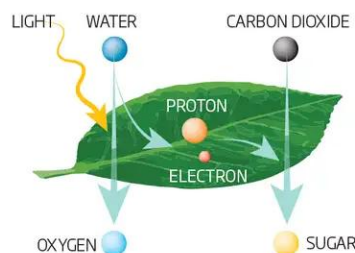
Green machines

A new generation of sunlight harvesters will be more useful than ever before



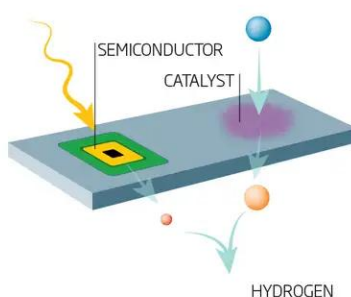
Natural leaf

During the day, plants take in water and carbon dioxide. They use light and a menagerie of enzymes to convert these into oxygen and sugar



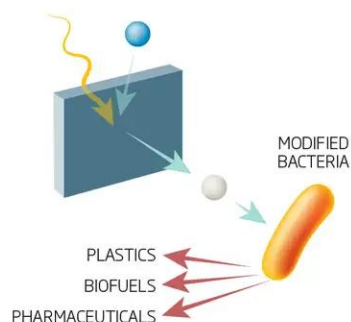
Artificial leaf

Synthetic leaves have a semiconductor to generate electrons from light, and a catalyst to steal protons from water. These are combined to make hydrogen



Bionic leaf

These combine light-harvesting tech with microbes. In one design, hydrogen from an artificial leaf is passed to microbes, which then produce useful chemicals



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BIRD FLYING INSPIRED MECHANISMS

- Modern mechanisms often involve morphing wings and tails for efficiency and maneuverability, seen in designs for drones and aircraft that mimic the bending and folding of bird wings

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GPS SYSTEM –A BIRD INSPIRED SYSTEM

- ideas from bird flying process is integrated in the GPS system
- Multi-sensor (sun, earth magnetic field and land mark) integration in birds is mimicked in GPS by utilizing signals from a network of satellites
- Birds adapt their flight paths based on environmental changes and GPS helps to reroute the paths based on the traffic congestion

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AIRCRAFT – A BIRD INSPIRED SYSTEM

- Aircraft design is heavily inspired by birds through biomimicry
- The body of the aircraft is modelled like the body of the bird , which minimizes air resistance
- The airfoils are shaped like wings of the bird, which facilitate the lift
- Similar to a bird's tail for control, aircraft use tail structures to manage altitude and direction.

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LOTUS LEAF EFFECT

- The Lotus Effect is a natural phenomenon observed on lotus leaves that provides them with a superhydrophobic, self-cleaning property

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LOTUS LEAF EFFECT

- Superhydrophobicity
 - The surface of a lotus leaf has a unique micro- and nanoscale roughness, with tiny bumps and waxy crystals. This structure traps air, minimizing the surface area of the leaf that meets water, making it highly water-repellent (superhydrophobic).
- Self-Cleaning
 - When water droplets land on the leaf, they maintain a spherical shape and roll off easily, picking up dirt and debris along the way. This rolling action effectively washes the leaf clean.

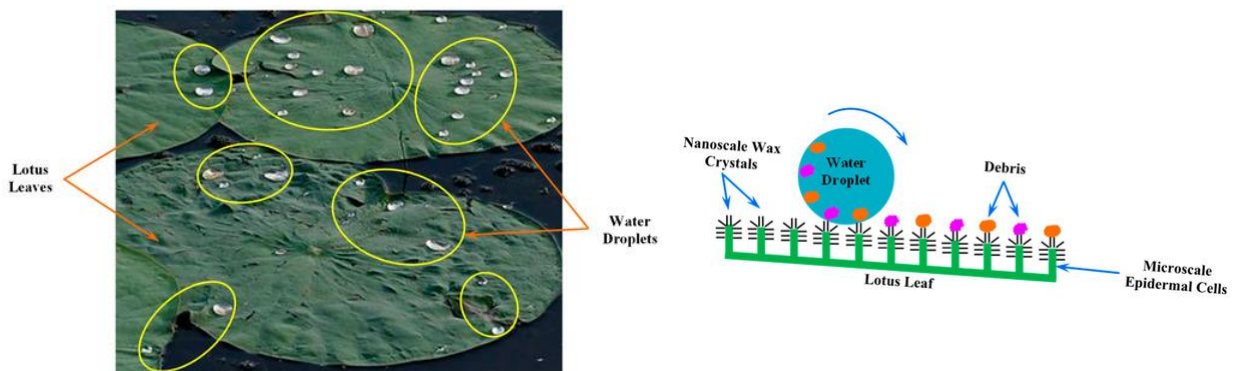
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LOTUS LEAF EFFECT



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APPLICATION OF LOTUS LEAF EFFECT

- Self-cleaning surfaces: Paints use this technology to keep buildings and other surfaces clean by washing away dirt with rain.
- Water-repellent fabrics: Coatings for textiles and upholstery that repel water and stains.
- Biomedical application : Self-cleaning and anti-fouling coatings for medical devices.
- Coatings for electronics: To protect electronic devices from water damage.

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PLAT BURRS

- The most distinctive feature of burrs are their numerous small hooks or teeth that act like a natural velcro.
- These hooks allow the burr to catch on anything with loops, such as animal fur or fabric, enabling the seeds inside to be transported to new locations for germination



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VELCRO

- Velcro was inspired by plant burrs, specifically burdock burrs, which feature tiny, microscopic hooks that easily attach to loops, such as fur or fabric.
- Consists of two parts
 - Tiny hooks similar to burr
 - Fabrics with loops and threads

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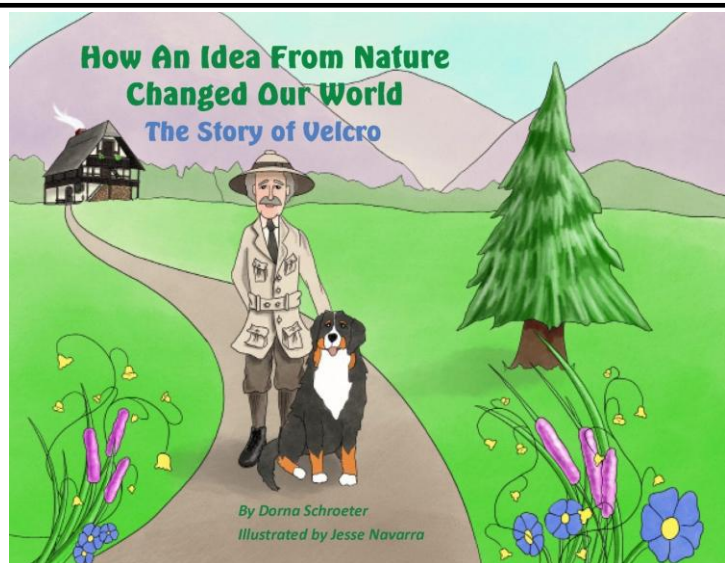
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STORY OF VELCRO

- Velcro was invented by Swiss engineer George de Mestral in the 1940s after he noticed how burrs from the burdock plant stuck to his dog's fur and his clothing during a hike



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SHARK SKIN

- Shark skin is covered in tiny, tooth-like scales called dermal denticles which reduce drag, allowing sharks to swim faster and more quietly



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FRICTION REDUCING SWIMSUITS IDEAS FROM SHARK SKIN

- Friction-reducing swimsuits inspired by shark skin incorporate microscopic, rib-like structures called dermal denticles to mimic the natural efficiency of shark skin.
- Speedo's Fastskin line and LZR Racer suit used this biomimicry, featuring textured fabrics that reduce drag by creating smaller, more orderly vortices in the water compared to a smooth surface



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KINGFISHER BEAK

- A kingfisher beak is a long, pointed, and streamlined tool that allows the bird to efficiently dive into water and catch prey without making a large splash or creating a significant compression wave.
 - The beak's gradually widening, cone-like shape helps the water flow smoothly around it, minimizing resistance.
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BULLET TRAIN NOSE INSPIRED BY KINGFISHER BEAK

- Engineers observed how kingfishers, with their long, pointed beaks, were able to dive into the water to catch fish with almost no splash.
- They realized this streamlined, biomimetic design could also reduce pressure waves and air resistance for the train.

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HUMAN BLOOD SUBSTITUTES

- Human blood substitutes like Hemoglobin-based oxygen carriers (HBOCs) and Perfluorocarbons (PFCs) are artificial oxygen-carrying solutions developed to avoid issues with blood supply and disease transmission from human transfusions

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HEMOGLOBIN-BASED OXYGEN CARRIERS (HBOCS)

- HBOCs use a modified version of the natural hemoglobin molecule found in red blood cells, which binds to oxygen.
- Challenges
 - Toxicity: Once removed from red blood cells (RBCs), hemoglobin can be toxic.
 - Adverse effects: Associated with increased blood pressure, cardiac issues, and damage to kidneys and other organs.
 - Vascular retention: Needs modification to prolong its presence in the bloodstream.
 - Supply: Relies on raw material from expired human blood, raising concerns about long-term availability

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PERFLUOROCARBONS (PFCS)

- These are synthetic molecules composed of carbon and fluorine that can dissolve and hold large amounts of oxygen, which is then released in low-oxygen environments.
 - Challenges
 - Emulsification: They are not soluble in water and require an emulsifier, such as albumin or lecithin, for intravenous use.
 - Particle size: Large particle size in older generations caused issues, including temporary drops in platelet count.
 - Persistence: The inert nature of PFCs can lead to them persisting in the body.
 - Discontinuation: Some earlier PFC-based products were withdrawn from the market due to limited success and side effects.
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END OF MODULE 3

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